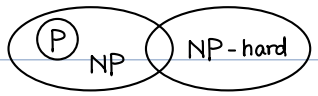
优化模型

$\min \|x\|_0 \quad s.t. \quad y = Ax \quad L_0\text{-min}$



0-norm 非凸 NP-hard

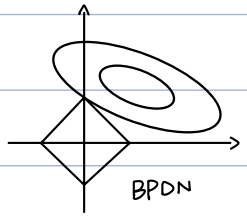
$\min \|x\|_1 \quad s.t. \quad y = Ax$ BP / L_1-min CVX, SPGL1

$\min \|x\|_1 \quad s.t. \quad \|y - Ax\| \leq \varepsilon$ BPDN

可行域 $\|y - Ax\| \leq \varepsilon$

$\|Ax^* - Ax\| = \|A(x^* - x)\|$ 椭圆

$Ax = y - \varepsilon$, $x = Null(A) + x_0$

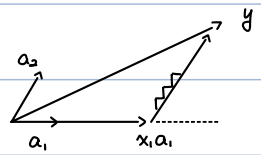


$\min \|y - Ax\|^2 + \lambda\|x\|_1$ LASSO CVX

LARS $n = L_1(g)$, move x_n toward $H_1(g) = A_n^H r$ until $L_1(g) \neq n$

$S = S \cup L_1(g)$, move x_S toward $A_S^H r$

min(M, N) Loops



$\min \|y - Ax\|^2 + \lambda\|x\|_1 + \delta\|x\|_2^2$ Elastic Net 变 Penalty

$\min \|y - Ax\|^2$ LS

闭式解 $\hat{x} = A^{\dagger}x = (A^H A)^{-1} A^H y$

$\min \|y - Ax\|^2 + \lambda\|x\|_2$ Ridge 适合 SNR 大

闭式解 $\hat{x} = (A^H A + \lambda I)^{-1} A^H y$

$\min \|y - Ax\|^2 + \lambda\|x\|_{TV}$ Trend Filtering (TV-norm)

$\min \|Y - AX\|^2 + \lambda\|X\|_{2,1}$ $L_{2,1}$ Group LASSO

$\min \|y - Ax\| + \lambda\|x\|_1$ SR-LASSO

SBL. MMSE. MAP 变 LOSS

Bregman. KL

$\min \|x\|_W \quad s.t. \quad y = Ax$ WBP

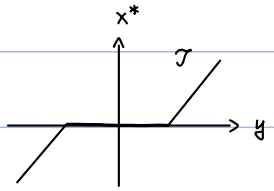
$$w_i = \begin{cases} \frac{1}{|x_i|}, & x_i \neq 0 \\ \infty, & x_i = 0 \end{cases}$$

$\min \|x\|_q \quad s.t. \quad y = Ax \quad L_q\text{-min}$

优化算法

LASSO　当 $A = I$ 时

$\min \ \frac{1}{2}\|y - x\|_2^2 + \lambda\|x\|_1$　　$x^* = \mathcal{T}_\lambda(y)$



Soft Threshold

$\min \ \frac{1}{2}(y - x)^2 + \lambda|x|$

$x^* = \mathcal{T}_\lambda(y)$　Soft Threshold

Pf: $x - y + \lambda \operatorname{sgn}(x) = 0$　　$y = x + \lambda \operatorname{sgn}(x)$

当 $y \geq \lambda$ 时，$x^* = y - \lambda$　　当 $y \leq \lambda$ 时，$x^* = y + \lambda$

当 $-\lambda < y < \lambda$ 时，$x^* = 0$

ADMM - BP

$\min \ \|x\|_1$　s.t. $y = Ax$

$\min \ I_C(x) + \|z\|_1$　s.t. $x = z$　　$C = \{x \mid y = Ax\}$

$L = I_C(x) + \|z\|_1 + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$

1) $x = [I - A^H(AA^H)^{-1}A](z - \frac{1}{\rho}\xi) + A^H(AA^H)^{-1}y$

Pf: $\min \ I_C(x) + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$

$\min \ \|x - \underbrace{z + \frac{1}{\rho}\xi}_{:=v}\|_2^2$　s.t. $y = Ax$

$L = \frac{1}{2}\|x - v\|_2^2 + \xi^H(y - Ax)$

$\frac{\partial}{\partial x}L = x - v - A^H\xi = 0$，$x = v + A^H\xi$ 左乘A

$y = Ax = Av + AA^H\xi$，$\xi = (AA^H)^{-1}(y - Av)$

$x = v + A^H(AA^H)^{-1}(y - Av) = [I - A^H(AA^H)^{-1}A]v + A^H(AA^H)^{-1}y$

2) $z = S_{1/\rho}(x + \frac{1}{\rho}\xi)$

Pf: $\min \ \|z\|_1 + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$

$\min \ \frac{2}{\rho}\|z\|_1 + 2 \cdot \langle \frac{1}{\rho}\xi, x - z \rangle + \|x - z\|_2^2$

$\min \ \|z - x - \frac{1}{\rho}\xi\|_2^2 + \frac{2}{\rho}\|z\|_1 - \cancel{\frac{1}{\rho^2}\|\xi\|_2^2}$

3) $\xi = \xi + \rho(x - z)$

ADMM

1) $x = \arg\min_x L$

2) $z = \arg\min_z L$

3) $\xi = \xi + \rho(x - z)$

ADMM - LASSO

$$\min \ \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|x\|_1$$

$$\min \ \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|z\|_1 \qquad \text{s.t. } x = z$$

$$L = \frac{1}{2}\|y - Ax\|_2^2 + \lambda\|z\|_1 + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$$

1) $x = (A^H A + \rho I)^{-1}(A^H y + \rho z - \xi)$

Pf: $\min \ \frac{1}{2}\|y - Ax\|_2^2 + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$

$$\frac{\partial}{\partial x} L = -A^H(y - Ax) + \xi + \rho(x - z) = 0$$

$$(A^H A + \rho I)x = A^H y + \rho z - \xi$$

2) $z = S_{\lambda/\rho}(x - \frac{1}{\rho}\xi)$

Pf: $\min \ \lambda\|z\|_1 + \langle \xi, x - z \rangle + \frac{1}{2}\rho\|x - z\|_2^2$

$$\min \ \frac{\lambda}{\rho}\|z\|_1 + \langle \frac{1}{\rho}\xi, x - z \rangle + \frac{1}{2}\|x - z\|_2^2 \qquad \times\frac{1}{\rho}$$

$$+ \langle \frac{1}{\rho}\xi, x - z \rangle - \frac{1}{\rho^2}\|\xi\|_2^2$$

$$\min \ \frac{\lambda}{\rho}\|z\|_1 + \frac{1}{2}\|x - z + \frac{1}{\rho}\xi\|_2^2 - \cancel{\frac{1}{\rho^2}\|\xi\|_2^2}$$

3) $\xi = \xi + \rho(x - z)$